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ORIGINAL ARTICLE

Malaria control at a gold mine in Sadiola District, Mali, and impact on transmission over 10 years

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Background: The SEMOS gold mine's malaria vector control programme forms part of the company's community responsibilities with the programme being managed by the mine's health department since 2005.

Methods: Data from approximately 10 years of malaria vector control for the Sadiola District are given: namely malaria vector control methods used by the control programme, positive malaria case data and entomological surveys from 2006, 2011 and 2014.

Results: Distribution of pyrethroid-treated bed nets and indoor residual spraying (IRS) with deltamethrin were implemented by the programme from 2005–2011. No IRS was done in 2012. Spraying with the organophosphate, pirimiphos-methyl resumed in 2013 and 2014 and was followed by a 70% drop in malaria cases in 2014. *Anopheles arabiensis* was the major vector present in 2006 and was susceptible to deltamethrin. In 2011 and 2014, *An. gambiae s.s.* was the most abundant vector with deltamethrin 24 h mortality of 68% and 19%, respectively.

Conclusions: Resistance to the pyrethroid deltamethrin has increased in *An. gambiae s.s.* since 2011, possibly due to the scale-up in distribution of long-lasting insecticide-treated bed nets. Resistance management strategies are recommended using different classes of insecticides for IRS, and including the distribution of new-generation bed nets.

Keywords: *Anopheles gambiae* complex, Insecticide resistance, Malaria transmission, Mali, Vector control

Introduction

Mali is divided into nine administrative regions (Kayes, Koulikoro, Sikasso, Ségou, Mopti, Gao, Tombouctou, Kidal and the capital, Bamako). Sadiola District falls within the Kayes administrative region in south-west Mali (Figure 1).

Mali national malaria control statistics reported by WHO[1] showed that distribution of insecticide-treated nets (ITNs) began on a national basis in 2005 and indoor residual spraying (IRS) was implemented in 2007. This was probably in response to the twofold increase in the number of reported suspected malaria cases and the number of deaths between 2001 and 2008.[1]

The SEMOS gold mine's malaria vector control programme was initiated by the mine's health department in 2005. The programme was started by the mine as a community based drive to aid in reducing mortality and morbidity amongst, firstly, mine employees and their dependents, and secondly, the surrounding villagers resident within the mine concession. The majority of the population relies on the mine for vector control IRS and on non-governmental organizations (NGOs) for the distribution of mosquito nets.

All four major malaria vector species occur in Mali: *Anopheles gambiae s.s., An. coluzzii, An. arabiensis* and *An. funestus*.[1–4] The first three are members of the *An. gambiae* complex and breed in temporary pools or rice paddies.[4] The first two are closely associated with humans and human habitations, while *An. arabiensis* is somewhat more variable and will readily feed on cattle and rest outdoors. *Anopheles funestus* is highly anthropophilic and endophilic and breeds in permanent, vegetated water bodies or slow-moving streams.[3]

This paper reports on malaria incidence and entomological surveillance following approximately 10 years of malaria vector

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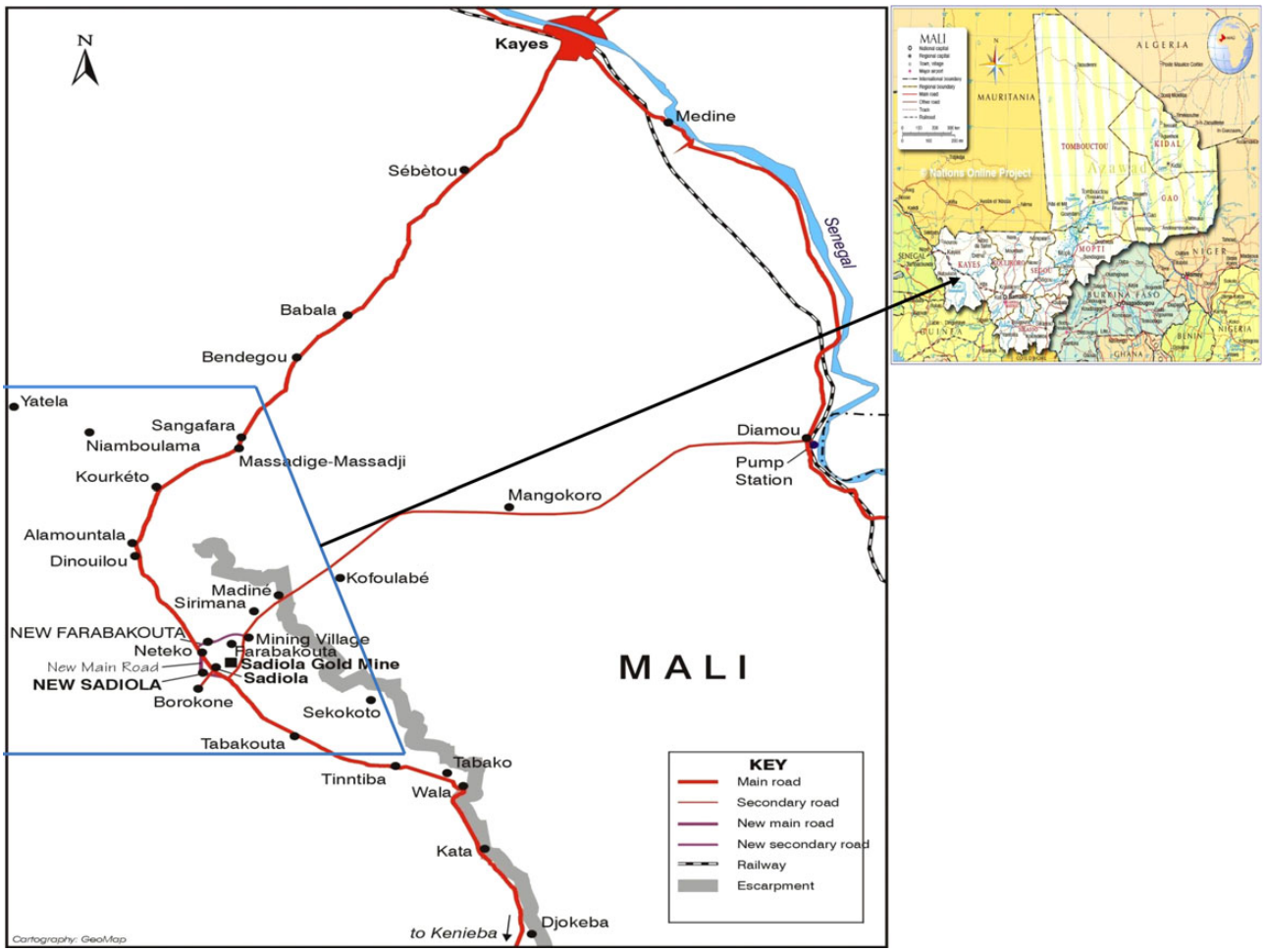


Figure 1. SEMOS mine spray zone (box) within the Sadiola District, Kayes administrative region, Mali. This figure is available in black and white in print and in colour at Transactions online.

control in the Sadiola District in south-west Mali by the SEMOS gold mine's vector control programme.

Materials and methods

Study area

The vector control programme covers an area of 396km^2, consisting of 17 villages, two mine camps SEMOS and Yatela, and the mines' work areas (Figure 1). Rainfall in the area is seasonal from May to October with a 10-year average rainfall of 836.8 mm. The area comprises an approximate human population of 26 282 based on malaria vector control data from 2014. Subsistence farming forms the basis of agricultural practices with rural or small-scale industry linked to those employed by the mine to aid in driving the local economy.

Malaria diagnosis and case management

Malaria diagnosis at the mines' clinics is carried out using the BIO-RAD OptiMAL-IT individual rapid diagnostic tests (RDTs). This is an immune-chromatographic test, using monoclonal antibodies against the metabolic enzyme pLDH (parasite lactate dehydrogenase) of *Plasmodium spp*. The antibodies can be classified into two groups, one specifically for *Plasmodium falciparum* and the other a pan-specific monoclonal antibody that reacts with all four species of *Plasmodium* occurring in humans, namely *P. falciparum*, *P. vivax*, *P. ovale* and *P. malariae*. The OptiMAL-IT test only gives a positive result when live parasites are present in the blood, hence it is suitable to ascertain the effectiveness of treatment and aid in identifying possible resistant strains of *Plasmodium* parasites. All malaria case data are based on positive cases treated at the mine clinics, being made up of the mine work force employees and dependents (contractors included) and does not include data from the local surrounding villages. The first-line treatment for uncomplicated malaria is artemisinin-based combination therapy.

Entomological surveys and insecticide susceptibility tests

Entomological surveys were carried out at the mine in September/October 2006, July/August 2011 and August 2014. All wild mosquitoes were collected resting indoors from households in villages inside and outside the spray area. The collections were either

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exposed to WHO insecticide-treated papers directly in the field (2006 and 2011), or transported to the Johannesburg laboratory for rearing and subsequent exposure of F-1 progeny (2014). Mosquitoes were exposed to all four classes of insecticide approved by WHO for use in vector control, namely organochlorines, organophosphates, carbamates and pyrethroids (see Tables 3–5 for a list of the insecticides tested), following the WHO standard protocol.[5] Results for the susceptibility tests were recorded 24 h post-exposure. The WHO criteria for susceptibility[5] were applied as follows: 98–100% mortality: susceptible; 90–97% requires further investigations; <90% mortality: resistance is confirmed. Species identification of the *Anopheles gambiae* complex and the *An. funestus* group was carried out using standard molecular methods.[6–8]

A subsample of the 2014 F-1 collection was used to carry out insecticide resistance intensity assays[9,10] using the time-response method. WHO 0.05% deltamethrin papers were used and two cohorts of 23 and 25 mosquitoes per tube were exposed continuously for 8 h with the WHO tubes lying horizontal. Knockdown was scored at 5, 10 and 20 min intervals up to 2 h and thereafter every hour up to and including 8 h total exposure time. The control colony used was a susceptible strain of *An. funestus* from Angola (FANG).

Vector control interventions

Currently, the SEMOS gold mine's malaria vector control programme consists of IRS, larviciding, active breeding site removal and household based malaria education. A summary of activities from 2005 to 2014 is presented in Table 1. The dipping and distribution of ITNs/long-lasting insecticide-treated nets (LLINs) by the mine was stopped at the end of 2011 after entomological studies revealed increased pyrethroid resistance in the local vectors.

The spray teams gather and compile data based on the number of houses sprayed, rooms per house, residents per house and

Table 1. Vector control interventions carried out at the SEMOS Gold Mine in south-western Mali

Year	Control method	Insecticide used	Coverage obtained
2005	IRS	Deltamethrin	Data unknown
	Net dipping	Deltamethrin: K-O Tab	
2006[a]	IRS	Deltamethrin	Data unknown
	Net dipping	Deltamethrin: K-O Tab	
2007	IRS	Deltamethrin	2862 houses sprayed
	Net dipping	Deltamethrin: K-O Tab	2417 nets dipped
			Population unknown
2008	IRS	Deltamethrin	4437 houses sprayed
	Net dipping	Deltamethrin: K-O Tab	2372 nets dipped
	Larviciding	*Bti* wettable granules	
2009	IRS	Deltamethrin	Data unknown
	Net dipping	Deltamethrin: K-O Tab	
	Larviciding	*Bti* wettable granules	
2010	IRS	Deltamethrin	997 houses sprayed
	Net dipping	Deltamethrin: K-O Tab	1173 nets dipped
	Larviciding	*Bti* wettable granules	Population unknown
2011[a]	IRS	Deltamethrin	4094 houses sprayed
	Net dipping	Deltamethrin: K-O Tab	Larviciding & net dipping data not available
	Larviciding	*Bti* wettable granules	Population = 18 746
2012	Larviciding	*Bti* wettable granules	872 water bodies treated and GPS plotted
	Breeding site removal		No IRS in 2012
	Education		Population unknown
2013	IRS	Pirimiphos-methyl: Actellic EC50	5293 houses sprayed
	Larviciding	*Bti* wettable granules	352 water bodies treated and GPS plotted
	Removal of breeding sites		Population 17 422
	Education		
2014[a]	IRS	Pirimiphos-methyl: Actellic EC50	7384 houses sprayed
	Larviciding	*Bti* wettable granules	205 water bodies treated
	Removal of breeding sites		Population 26 282
	Education		

Bti: *Bacillus thuringiensis* var. *israelensis*; IRS: indoor residual spraying.
[a] The years in which entomological surveys were carried out.

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the number of nets found in each house. The data are captured on a daily basis using spreadsheets for analysis.

Larviciding with *Bacillus thuringiensis* var. *israelensis* (*Bti*) was carried out extensively in 2012 but scaled back in subsequent years due to low rainfall.

A gap year in IRS occurred in 2012 due to political instability in Mali as a result of a military coup. With this change in the government, the use of the granular based wettable organophosphate, pirimiphos-methyl, was prohibited as this formulation of the insecticide was not on the Malian governments' promulgated list. A liquid-based formulation of the same insecticide–Actellic EC50® was only able to reach the site at the end of the vector control season, thus IRS was only done on a small scale involving only the work place. From 2013 to 2014 IRS was reintroduced into the vector control program and was carried out aggressively in the area.

Results

Malaria case data

Cases recorded by the mine health department in the SEMOS mine clinics from 2004 to 2014 are given in Figure 2. Limited data were available for malaria cases in children under the age of 16 years and for population estimates prior to 2011, as previous years were not recorded. From 2004 until 2008 positive cases were based on either clinical presentation or a positive RDT. From 2009 only RDT confirmed positive cases resulted in malaria treatment being given. The number of malaria positive cases ranged from a low of 289 in 2006 to 1794 in 2012. Malaria positive cases were further broken down to compare total cases to those of patients under the age of 16 years. This equated to 34.4% in 2011, 32.7% in 2012, 48.8% in 2013 and 54.6% in 2014.

Entomological surveys

Three members of the *An. gambiae* complex were identified[6,7] from the SEMOS mine area: *An. gambiae*, *An. coluzzii* and *An. arabiensis* (Table 2). In 2006, the dominant species was *An. arabiensis* (92%) but this changed in 2011 and 2014 with *An. gambiae s.s.* becoming the dominant species (80 and 70% respectively). *Anopheles coluzzii* was rare in all three survey years. In 2006, 11 specimens of *An. funestus*[8] were collected followed by 10 specimens in 2011. Seven specimens of *An. rivulorum*[8] were collected in 2006. No *An. funestus* group mosquitoes were collected in 2014 (Table 2).

The insecticide susceptibility test results are shown in Tables 3–5 for all three surveys and all classes of insecticides that were tested.[5] In 2006 (Table 3), a single *An. coluzzii* was resistant to DDT and a single *An. gambiae s.s.* was resistant to bendiocarb. The one specimen that survived exposure to deltamethrin was a member of the *An. gambiae* complex but not identified to species. Of 58 *An. arabiensis* tested on bendiocarb, mortality was 56.9%. In 2011 (Table 4), of the 41 survivors on deltamethrin, 36 were *An. gambiae s.s.*, two *An. coluzzii* and three *An. arabiensis*. Of 90 survivors on the two carbamates, 76 were *An. gambiae s.s.*, two *An. coluzzii* and 12 *An. arabiensis*. On DDT, 28 *An. gambiae s.s.* and one *An. arabiensis* survived exposure. In 2014, all adult mosquitoes were returned to the Johannesburg laboratory, induced to

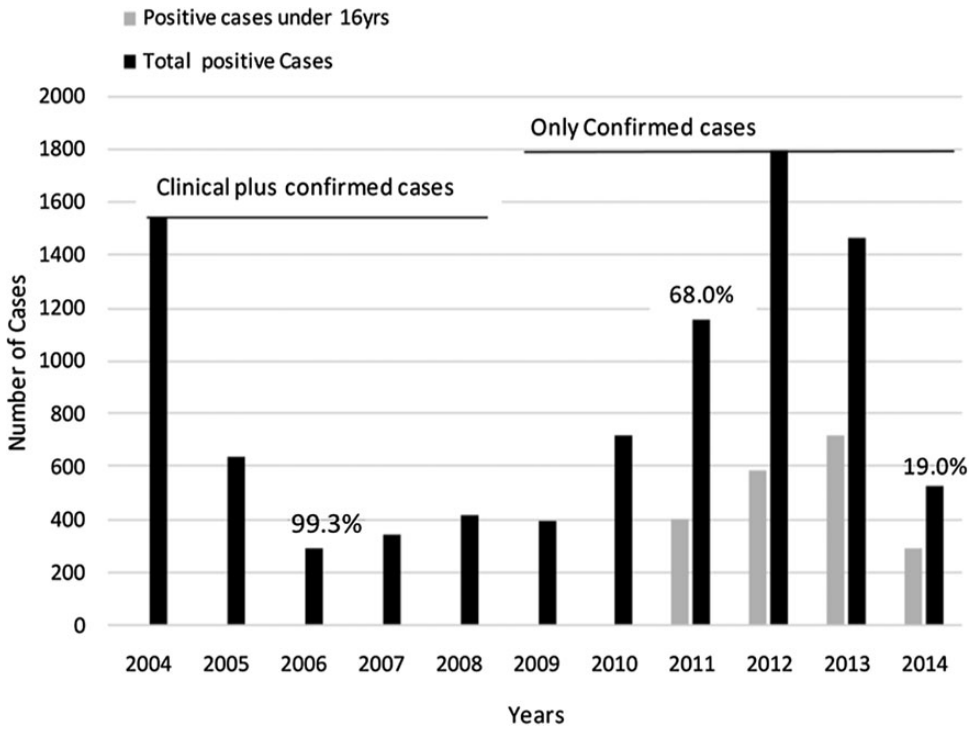


Figure 2. Malaria positive cases by year recorded by the mine health department in the SEMOS mine clinics from 2004 to 2014. Percentages given above 2006, 2011 and 2014 are the 0.05% deltamethrin 24 h post-exposure mortalities recorded for *Anopheles arabiensis* in 2006, and *An. gambiae s.s.* in 2011 and 2014.

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Table 2. Species identification of wild *Anopheles* material collected from the SEMOS Gold Mine in the Sadiola District, Mali, using standard PCR methods

Year	Species Identified	No.
2006	*An. gambiae s.s.*	6
22 September to 5 October	*An. coluzzii*	6
	An. arabiensis	147
	An. funestus	11
	An. rivulorum	7
2011	*An. gambiae s.s.*	196
25 July to 6 August	*An. coluzzii*	5
	An. arabiensis	43
	An. funestus	10
2014	*An. gambiae s.s.*	165[a]
4 to 17 August	*An. coluzzii*	15[a]
	An. arabiensis	57[a]

[a] Number of wild females that laid eggs in the Johannesburg laboratory and whose progeny was used for the insecticide susceptibility and intensity tests.

Table 3. Pooled insecticide resistance tests carried out in the field at the SEMOS Gold Mine in 2006, where 92% of 159 samples were identified by PCR as *Anopheles arabiensis* (Table 2)

Insecticide	No. tested	% Mortality 24 h post-exposure	Status[a]
Deltamethrin 0.05%	134	99.3	Susceptible
Malathion 5.0%	94	100	Susceptible
Bendiocarb 0.1%	81	74.0	Resistant
DDT 4.0%	104	94.2	Resistant
Dieldrin 4.0%	123	74.8	Resistant
Controls[b]	48	2.1	NA

NA: not applicable.
[a] Resistant or susceptible according the current WHO criteria.[5]
[b] Controls consisted of exposures to untreated papers.

lay eggs and the susceptibility tests carried out on identified F-1 progeny (Table 5). The decrease in deltamethrin mortality is mirrored by a decrease in DDT mortality. The low mortalities seen on the two carbamates in 2011 was reversed in 2014. Figure 2 gives the overall percent mortality 24 hours post-exposure to 0.05% deltamethrin for the three survey years.

The insecticide intensity tests (Figure 3) showed that the Sadiola *An. gambiae s.s.* population was highly resistant to deltamethrin with only 50% mortality after 8 h exposure to WHO treated papers compared with the *An. funestus* susceptible strain (FANG) where 100% mortality was achieved within 1 hour.

Table 4. Pooled insecticide resistance tests carried out in the field at the SEMOS Gold Mine in 2011, where 80% of the identified specimens (Table 2) were *Anopheles gambiae s.s.*

Insecticide	No. tested	% Mortality 24 hrs post-exposure	Status[a]
Deltamethrin 0.05%	125	68.0	Resistant
Malathion 5.0%	96	96.0	Resistant
Fenitrothion 1.0%	83	98.8	Susceptible
Bendiocarb 0.1%	59	11.9	Resistant
Propoxur 0.1%	48	20.8	Resistant
DDT 4.0%	47	38.3	Resistant
Dieldrin 4.0%	17	82.4	Resistant
Controls[b]	63	0	NA

NA: not applicable.
[a] Resistant or susceptible according the current WHO criteria.[5]
[b] Controls consisted of exposures to untreated papers.

Vector control activities

Data gathered from the vector control rounds in 2011, 2013 and 2014, showed that the number of bed nets per household to houses sprayed increased in the IRS spray area (Figure 4). Despite the mine programme having stopped all activities directed at dipping or distribution of nets in 2011, there was a higher number of nets in households after this was stopped than before. The dramatic increase in the local human population numbers (from an estimated 6097 in 2010 to 26 282 in 2014) could be an artefact of better IRS data capturing by the spray teams rather than a real four-fold increase in the human population over 5 years.

In 2012, the malaria vector control programme concentrated on mosquito breeding site control only in the form of weekly larviciding with *Bti* and GPS plotting 872 water bodies for future reference. In 2013 and 2014 fewer water bodies were treated (Table 1) due to low rainfall. Intensive education campaigns engaging local village elders was started and aimed at the removal of breeding sites and good housekeeping practices.[12]

Discussion

Vector control

The SEMOS programme from 2004–2011 followed the prescribed vector control methods,[13] with IRS and distribution of ITN/LLINs forming the major drive, and larviciding only occurring late in the spray season. This line of control continued with only one mosquito survey being carried out during these years (in 2006). When the number of cases increased almost three-fold from 2009 to 2011, a second mosquito survey was commissioned. This survey (in 2011) showed a marked increase in insecticide resistance to pyrethroids, resulting in all use of pyrethroid insecticides in the SEMOS vector control programme being stopped, including net dipping and net distribution.

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Table 5. Insecticide resistance tests carried out in 2014 on 2–5 day old F-1 female progeny of wild *Anopheles* mosquito females identified by PCR

Insecticide	Species	No. tested	% Mortality 24 hrs post-exposure	Status[a]
Deltamethrin 0.05%	*An. gambiae s.s.*	58	19.0	Resistant
	An. arabiensis	28	85.7	Resistant
Pirimiphos-methyl 0.25%	*An. gambiae s.s.*	100	100	Susceptible
	An. arabiensis	85	98.8	Susceptible
	An. coluzzii	28	100	Susceptible
Bendiocarb 0.1%	*An. gambiae s.s.*	109	93.6	Resistant
	An. arabiensis	105	83.8	Resistant
DDT 4.0%	*An. gambiae s.s.*	48	2.1	Resistant
Controls[b]	*An. gambiae s.s.*	50	2.0	NA

NA: not applicable.
[a] Resistant or susceptible according the current WHO criteria.[5]
[b] Controls consisted of exposures to untreated papers.

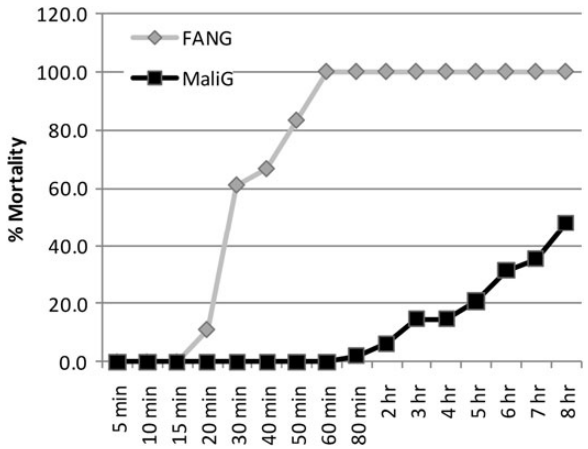


Figure 3. Resistance intensity tests of *An. gambiae s.s.* from Mali (MaliG) exposed for 8 h to 0.05% deltamethrin in a WHO tube test. FANG: insecticide susceptible *An. funestus* laboratory strain.

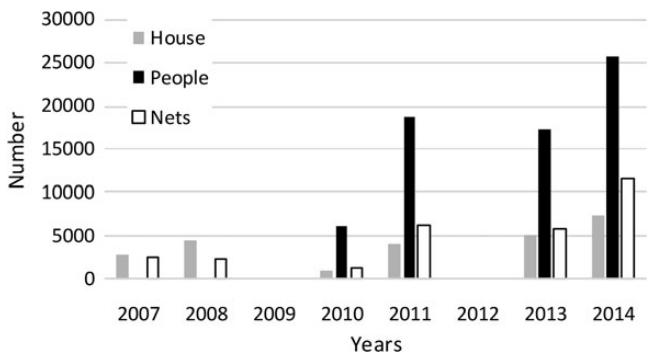


Figure 4. The number of total population covered, households sprayed and mosquito nets dipped and distributed from 2007–2014 in the Sadiola District spray zone. No houses were sprayed during 2012 owing to political instability.

Based on the 2011 entomological survey, the IRS insecticide was changed to the organophosphate pirimiphos-methyl as a means of managing the resistance problem in accordance with the WHO Global Plan for Insecticide Resistance Management.[14] However, due to the military coup in Mali in 2012, no IRS was carried out that year. The concomitant rise in malaria cases to 1794 cannot be ascribed to over-diagnosis since all cases were confirmed by RDT, as in the previous 3 years. Access to the health system remained unchanged and there was no indication of treatment failure indicating that drug resistance might be the cause of the increase. Without the use of IRS with an effective insecticide, the resistant vector populations would have increased despite larviciding being carried out. It is, therefore, not surprising that an increase in malaria case numbers was seen. Once the IRS was reintroduced, a 70% reduction in malaria cases was recorded within two years, demonstrating the effectiveness of this vector control tool.

Nets per household in 2014 equated to an average of 1.59 nets compared to previous data gathered from 2013 where the average number of nets per household was 1.07. This was a result of the President's Malaria Initiative (Malaria Operational Plan FY 2014),[11] which stated the need for increased distribution of treated nets as part of the universal coverage campaign and routine distribution through antenatal care (ANC) and child immunization clinics. Mali defines achievement of universal coverage as one LLIN for every two persons.[11]

Positive malaria cases

The malaria cases reported from 2004 to 2008 in Figure 2 are a combination of clinically diagnosed and RDT confirmed positive cases. As in most cases in rural malaria areas, treatment for malaria is often given without positive malaria test results, leading to possible misdiagnoses of other medical conditions such as meningitis, dengue, typhoid and other conditions where pyrexia is common.[15] In 2009 the mine's department of health implemented a policy whereby only a positive malaria test would result in

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anti-malaria treatment being started. Thus all numbers reflected in Figure 2 from 2009 to 2014 are confirmed positive for malaria. This policy implementation allowed for cases of dengue fever in the area to be diagnosed in 2010.

Malaria case data from 2009–2012 showed an increase in the number of cases associated with an increase in insecticide resistance in the vector mosquitoes. The year 2012 saw the highest number of malaria cases recorded which was coincident with no IRS activity, and although other malaria vector control methods were undertaken during this year, they had no impact on transmission. By the close of the 2014 malaria transmission season, a 70% reduction in positive malaria cases was seen, from 1794 cases in 2012 down to 527 in 2014 (Figure 2).

Entomological surveys

The change in species composition from 92% *An. arabiensis* in 2006 to 70–80% *An. gambiae s.s.* in 2011/2014 is interesting. This could have been due to climatic conditions becoming more favourable for *An. gambiae s.s.* or the rise of insecticide resistance in this species conferring a selective advantage over *An. arabiensis*. The presence of low numbers of *An. coluzzii* could be because their preferred breeding habitats are not available, these being mainly large bodies of water associated with rice paddies.[16]

The continued increase in resistance to the pyrethroid deltamethrin is alarming as the SEMOS mine malaria vector control programme stopped using deltamethrin at the end of 2011. This included all dipping and distribution of bed nets, as well as a 'gap' year of no IRS in 2012 due to the political situation in the country. The same period saw a marked increase in DDT resistance presumably as a result of cross-resistance between pyrethroids and DDT based on mutations at the shared target site in the sodium channel gene.[17] The results from an 8 h exposure of *An. gambiae s.s.* to the WHO diagnostic dose of 0.05% deltamethrin in 2014, giving only 50% mortality, are of grave concern.

Pyrethroid insecticide resistance has been well documented throughout malaria-affected countries in Africa.[18,19] It is inevitable that insecticide resistance was going to have a direct effect on re-emerging vector borne diseases and where resistance had not developed, it was expected to threaten disease control in the future.[18] Pyrethroid resistance in *An. gambiae* was first reported in Cote d'Ivoire as early as 1998[20] and resistance is now widespread throughout the African continent. The use of pyrethroids for both IRS and ITNs and their relationship to the developing pyrethroid resistance has been documented by a number of authors from a variety of countries throughout Africa,[21–27] and as early as 2007[28] the efficacy of the pyrethroid-treated bed nets was being questioned. There is a growing body of information supporting the hypothesis that the use of pyrethroids without an effective resistance management programme[14] has in fact exacerbated the insecticide resistance problem in Africa. Continued and expanded use of pyrethroids on fabrics and as stand-alone residual insecticides for IRS can only result in the elimination of the most useful insecticide currently available to us. The use of single products in control programmes (as was done here with pyrethroids from 2005–2011) is likely to result in high levels of resistance to more of the very few molecules available to us. The latest report from the USAID/President's Malaria Initiative on insecticide resistance in the countries where they operate, including Mali, indicates the seriousness of the situation with most mosquito populations showing resistance to three classes of insecticide: pyrethroids, organochlorines and carbamates.[29]

Limitations of the study

This paper reports on the activities of an operational malaria control programme in south-western Mali and is not a structured research project with clearly defined objectives. The sole objective of the control programme is to reduce the burden of malaria. As a result, there are gaps in the data over the years as programme staff have moved on. If the entomological surveys had been carried out more regularly, some of the questions regarding change in species composition and evolution of resistance may have been answered. Nevertheless, the data presented here certainly seems to indicate that the resistance had an impact on transmission and that this only improved when an effective insecticide was used for IRS.

Conclusions

A change in vector control policy by the SEMOS malaria control programme in the face of pyrethroid resistance saw a 70% decrease in malaria cases between 2012 and 2014. Despite the change to pirimiphos-methyl, the resistance to pyrethroids in *An. gambiae s.s.* has increased dramatically, and so too has the treated net distribution within the Sadiola spray zone by various NGO's. This highlights the need for alternative, novel vector control interventions that will assist control programmes to manage the resistance and still control transmission adequately. It is encouraging, therefore, that many studies are being supported to develop new insecticides for IRS and new technology for LLINs.

Authors' disclaimer: The views expressed in this paper are those of the authors and cannot be used against SEMOS or its shareholders in any form, but are intended for study purposes only.

Authors' contributions: The SEMOS mine department of health personnel (SW, DT, MarC) are responsible for the implementation of the malaria control programme. AG, RC, GS, RH and MauC were responsible for carrying out the entomological surveys, laboratory analysis and resistance intensity tests. SW drafted the initial manuscript and all authors contributed to and approved the final submission. MauC is the guarantor of the paper.

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Ethical approval: Not required.

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